第1题. 写为增广矩阵形式.

$$\begin{bmatrix} 1 & 4 & 0 & -1 & -1 \\ 1 & -2 & 2 & 3 & 3 \\ 3 & 6 & 2 & 2 & 3 \\ 1 & -8 & 4 & 7 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & 0 & -1 & -1 \\ 0 & -6 & 2 & 4 & 4 \\ 3 & 6 & 2 & 2 & 3 \\ 1 & -8 & 4 & 7 & 7 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 4 & 0 & -1 & -1 \\ 0 & -6 & 2 & 4 & 4 \\ 0 & -6 & 2 & 5 & 6 \\ 1 & -8 & 4 & 7 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & 0 & -1 & -1 \\ 0 & -6 & 2 & 4 & 4 \\ 0 & -6 & 2 & 5 & 6 \\ 0 & -12 & 4 & 8 & 8 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 4 & 0 & -1 & -1 \\ 0 & -6 & 2 & 4 & 4 \\ 0 & -6 & 2 & 5 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & 0 & -1 & -1 \\ 0 & -6 & 2 & 4 & 4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 4 & 0 & -1 & -1 \\ 0 & -3 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & 0 & -1 & -1 \\ 0 & -3 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 4 & 0 & 0 & 1 \\ 0 & -3 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

得 $x_1 + 4x_2 = 1$

$-3x_2 + x_3 = -2$

$x_4 = 2$

所以 $\begin{cases} x_1 = \dfrac{-5-4x_3}{3} \\ x_2 = \dfrac{2+x_3}{3} \\ x_3 \text{为自由变量}. \\ x_4 = 2. \end{cases}$

2. $\left[\begin{array}{cccccc} 1 & 0 & -2 & 1 & 0 & 0 \\ -3 & 1 & 4 & 0 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{array}\right]$

$\sim \left[\begin{array}{cccccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{cccccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & -3 & 8 & -2 & 0 & 1 \end{array}\right]$

$\sim \left[\begin{array}{cccccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{array}\right] \sim \left[\begin{array}{cccccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{array}\right]$

$\sim \left[\begin{array}{cccccc} 1 & 0 & 0 & 8 & 3 & 1 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{array}\right]$

$\sim \left[\begin{array}{cccccc} 1 & 0 & 0 & 8 & 3 & 1 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 1 & \frac{7}{2} & \frac{3}{2} & \frac{1}{2} \end{array}\right]$ 得 $A^{-1} = \left[\begin{array}{ccc} 8 & 3 & 1 \\ 10 & 4 & 1 \\ \frac{7}{2} & \frac{3}{2} & \frac{1}{2} \end{array}\right]$

3. $X = \left[\begin{array}{c} x_1 \\ x_2 \\ x_3 \end{array}\right]$ $A = \left[\begin{array}{ccc} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 4 & 9 & 16 \end{array}\right]$ $b = \left[\begin{array}{c} 1 \\ -1 \\ 1 \end{array}\right]$

$Ax = b$; $x_i = \dfrac{\det A_i(b)}{\det A}$

$\det A = 2$.

$x_1 = \frac{\det A_1(b)}{\det A} = \frac{\begin{vmatrix} 1 & 1 & 1 \\ -1 & 3 & 4 \\ 1 & 9 & 16 \end{vmatrix}}{2} = \frac{20}{2} = 10$

$x_2 = \frac{\det A_2(b)}{\det A} = \frac{\begin{vmatrix} 1 & 1 & 1 \\ 2 & -1 & 4 \\ 4 & 1 & 16 \end{vmatrix}}{\det A} = \frac{30}{2} = -15$

$x_3 = \frac{\det A_3(b)}{\det A} = \frac{\begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & -1 \\ 4 & 9 & 1 \end{vmatrix}}{\det A} = \frac{12}{2} = 6$

即 $X = \begin{bmatrix} 10 \\ -15 \\ 6 \end{bmatrix}$ 即 $x_1 = 10$, $x_2 = -15$, $x_3 = 6$

4. ~~AA~~ B

(1)

(2) $A^n - 2A^{n-1} = (A-2)(A^n - A^{n-1})$

$= (A-2)(A^{n-1})$

$= (A - 2I)(A^{n-1}) = \left(\begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \right) \cdot A^{n-1}$

$$= \begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{bmatrix} \cdot A^{n-1}$$

$$= \begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{bmatrix} \cdot A \cdot A^{n-2}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot A^{n-2} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$$

5. (1) 证明:

$$(kA)^{-1} = \left(k \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \right)^{-1}$$

$$= \left(\begin{bmatrix} ka_{11} & ka_{12} & \cdots & ka_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ ka_{n1} & ka_{n2} & \cdots & ka_{nn} \end{bmatrix} \right)^{-1}$$

要求逆 $\begin{bmatrix} ka_{11} & ka_{12} & \cdots & ka_{1n} & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & 0 & \vdots & \ddots & \vdots \\ ka_{n1} & ka_{n2} & \cdots & ka_{nn} & 0 & 0 & \cdots & 1 \end{bmatrix}$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & \frac{1}{k} & \cdots & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & 0 & \ddots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & 0 & 0 & \cdots & \frac{1}{k} \end{bmatrix} \quad ①$$

$a_{1n}\ a_{2n}$

因 $$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & 1 & \cdots & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} & 0 & 0 & \cdots & 1 \end{bmatrix} \quad ②$$

由②求出 A 的逆为 A^{-1}

由①同理 $(kA)^{-1}$ 的逆为 $\frac{1}{k}\cdot A^{-1}$

(2) 证明: $AB = A+B$

$$A^{-1}\cdot AB = A^{-1}\cdot(A+B)$$

$$B = I + A^{-1}\cdot B$$

$$B\cdot B^{-1} = (I + A^{-1}\cdot B)\cdot B^{-1}$$

$$I = B^{-1} + A^{-1}$$

$$AB = ABI = AB\cdot(B^{-1}+A^{-1})$$

$$= A + ABA^{-1}$$

~~$BA = IBA = (B^{-1}+A^{-1})\cdot BA$~~

~~$= A+I$~~

$AB = IAB = (B^{-1}+A^{-1})AB = B + B^{-1}AB$.

因为 $AB = A+B$. 所以 $A+B = B+B^{-1}AB$. 即 $A = B^{-1}AB$

$BA = BAI = BA(B^{-1}+A^{-1}) = BAB^{-1} + B$

$= B(B^{-1}AB)B^{-1} + B$

$= A + B = AB$.

所以证得 $AB = BA$

第6题

能.

证明：设 $c_1, c_2, c_2 \cdots c_r$ 为非全为0的标量

$\beta = c_1a_1 + c_2a_2 + c_3a_3 + \cdots c_ra_r$

若 $c_r = 0$. 则 $\beta = c_1a_1 + c_2a_2 + \cdots c_{r-1}a_{r-1}$

由题知. β 不能由 $a_1, a_2 \cdots a_{r-1}$ 线性表示

则与题矛盾. 所以 $c_r \neq 0$.

$\beta - c_1a_1 - c_2a_2 - \cdots c_{r-1}a_{r-1} = c_ra_r$.

$$a_r = \frac{\beta - c_1a_1 - c_2a_2 - \cdots - c_{r-1}a_{r-1}}{c_r}$$

所以 a_r 可以由 $a_1, a_2 \cdots a_{r-1}$ 线性表示

第七题，每一行各减第一行得，且行列各为n

$$A=\begin{bmatrix} a & b & b & \cdots & b \\ b-a & a-b & 0 & \cdots & 0 \\ b-a & 0 & a-b & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b-a & 0 & 0 & \cdots & a-b \end{bmatrix} = (a-b)^{n-1} \begin{bmatrix} a & b & b & \cdots & b \\ -1 & 1 & 0 & \cdots & 0 \\ -1 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

每一列加到第一列得

$$(a-b)^{n-1} \begin{bmatrix} a+(n-1)b & b & b & b & b \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

展开得 $(a-b)^{n-1} \cdot [a+(n-1)b] \cdot \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$= (a-b)^{n-1} \cdot [a+(n-1)b]$$

第8题 证 因为A是2×2矩阵，因C右乘A，

则C有2列

因C左乘A，则C有2行

所以C为2×2矩阵.

设 $C_1=\begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix}$.

$$AC=\begin{bmatrix} 1 & a \\ 1 & 0 \end{bmatrix}\begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix}=\begin{bmatrix} c_1+ac_3 & c_2+ac_4 \\ c_1 & c_2 \end{bmatrix}$$

$$CA=\begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix}\begin{bmatrix} 1 & a \\ 1 & 0 \end{bmatrix}=\begin{bmatrix} c_1+c_2 & ac_1 \\ c_3+c_4 & ac_3 \end{bmatrix}$$

则 $AC-CA=\begin{bmatrix} ac_3-c_2 & c_2+ac_4-ac_1 \\ c_1-c_3-c_4 & c_2-ac_3 \end{bmatrix}$

$AC-CA=B$ 则 $\begin{bmatrix} ac_3-c_2 & c_2+ac_4-ac_1 \\ c_1-c_3-c_4 & c_2-ac_3 \end{bmatrix}$

$$=\begin{bmatrix} 0 & 1 \\ 1 & b \end{bmatrix}$$

则 $\begin{cases} ac_3-c_2=0 & ① \\ c_2+ac_4-ac_1=1 & ② \\ c_1-c_3-c_4=1 & ③ \\ c_2-ac_3=b & ④ \end{cases}$

①+④得 $b=0$ 则 $c_2=ac_3$

$ac_3+ac_4-ac_1=1$ ⑤

~~$c_1=c_3+c_4$~~ ③=⑤.

则 $a=-1$.

所以 $a=-1$, $b=0$